

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8629**

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	409	Seat No:	2022700008
Course Section:	1				

L_32_631_986_L_1505_41422



Name of the candidate. Keya Dey

Candidate's UID number:

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Examination class room number:

4	0	9
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Invigilator's signature with date:

 08/10/25

(To be filled in by the candidate)
EXAMINATION: B.E. Btech
(For example FY MCA/FY BTECH)

PROGRAM: CSE - DS

DAY AND DATE: 8th Oct '25

COURSE NAME: Deep Learning

SEMESTER :

I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.


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For
Marks

 Question
No.

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(Begin answer for each question on a new page)

Q1.

$$a. f(w) = \frac{1}{2} \|x_w - y\|^2$$

Gradient update rule:

$$\theta \rightarrow \theta - n \cdot f(w) \cdot n \cdot \Delta L$$

$$\rightarrow \theta - n \cdot f(w)$$

$$\rightarrow \theta - n f'(w)$$

$$\rightarrow \theta - n \frac{d(f(w))}{d(x)}$$

$$\rightarrow \theta - \frac{n}{2} \frac{d(\|x_w^2 - 2x_w y + y^2\|)}{dx}$$

$$\rightarrow \theta - \frac{n}{2} (2x_w' - 2y + y^2)$$

$$W = W_{old} - n \cdot (x_w' - y + y^2)$$

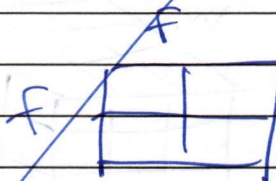
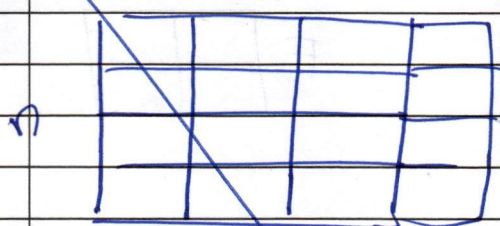
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Q2

a. Standard 2-D convolution:



no. of elements

size of output

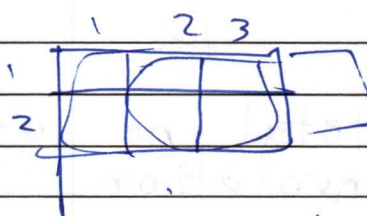
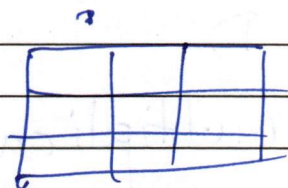
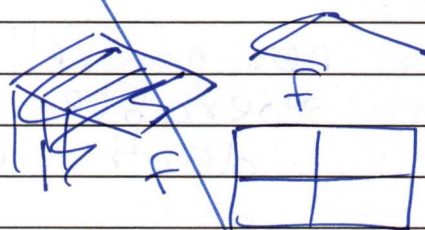
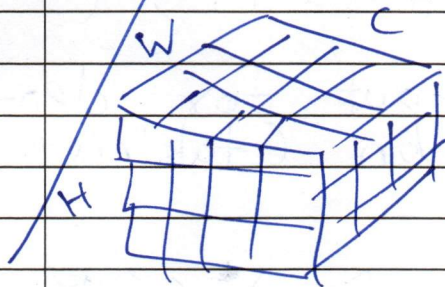
$$= (n - f + 2p) + 1$$

$$= n - f + 1$$

$\therefore$ total no. of operations
(f ~~oper~~ multiplications per box)

$$= f * (n - f + 1)$$

Depthwise separable convolution:



$$2 \times 2$$

$$3 - 2 + 1$$

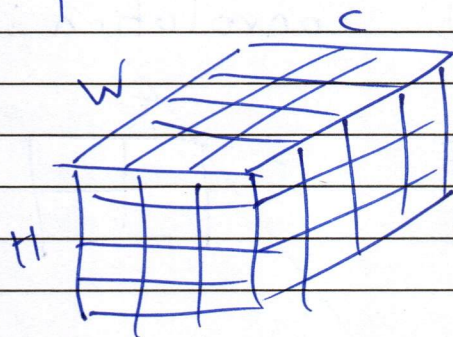
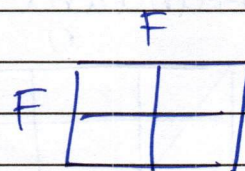
$$4 - 2 + 1$$

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Q2a. Input :


 s
 4
 2


standard 2D convolution no. of cells in output :

$$\cancel{H \times W} = (W - F + 1) * (H - F + 1)$$

$\therefore$ ~~no. of~~ each cell has been derived with $F * F$ multiplications

$\therefore$ total no. of multiplications for convolution

$$= (W - F + 1) * (H - F + 1) * (F * F) \quad \text{--- (1)}$$

no. of cells in output for depthwise separable convolution (taking depth as D)

$$= \frac{(W - F + 1)}{D} * \frac{(H - F + 1)}{D} * (F * F)$$

$\therefore$ total no. of multiplications for convolution

$$= \left(\frac{W - F + 1}{D} \right) * \left(\frac{H - F + 1}{D} \right) * (F * F)$$

$$= \frac{(W - F + 1)}{D} * \frac{(H - F + 1)}{D} * F^2 \quad \text{--- (2)}$$



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Q2.

a. $\therefore$ depthwise separable convolution requires lesser no. of multiplications than standard 2D convolution by a factor of $\frac{1}{D^2}$

① no. of multiplications in depthwise sep. convolution = $\frac{1}{D^2} \times$ no. of multiplications in standard convolution.

② $\therefore$ it is much smaller than the standard.

③ Due to its ~~not~~ being a lot more smaller, when input sizes get too large, depthwise separable convolution ~~shows~~ will show much better performance.

④ It reduces the no. of parameters by a large margin, making it parameter efficient.

⑤ It is therefore critical in modern architectures.



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Q2.

b. Subspace learnt by PCA

$$= \begin{vmatrix} \text{cov}(X_1, X_1) & \text{cov}(X_1, X_2) \\ \text{cov}(X_2, X_1) & \text{cov}(X_2, X_1) \end{vmatrix}$$

$$\text{cov}(X_1, X_2) = \frac{1}{n-1} \sum_{i=0}^n (\bar{X}_1 - X_{i1})(\bar{X}_2 - X_{i2})$$

$$\text{MSE} = \frac{1}{n} \sum_{i=0}^n (\bar{X} - X_{i1})^2$$

eq - subspace learnt by linear undercomplete autoencoder trained with MSE for X_1, X_2

$$= \frac{1}{n} \sum_{i=0}^n (\bar{X}_1 - X_{i1})(\bar{X}_2 - X_{i2})$$

both subspaces can be generalised as

$$\frac{1}{k} \sum_{i=0}^n (\bar{X}_1 - X_{i1})(\bar{X}_2 - X_{i2})(\bar{X}_3 - X_{i3}) \dots (\bar{X}_n - X_{in})$$

(k is constant)

Thus, both subspaces are equivalent.



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Q2.

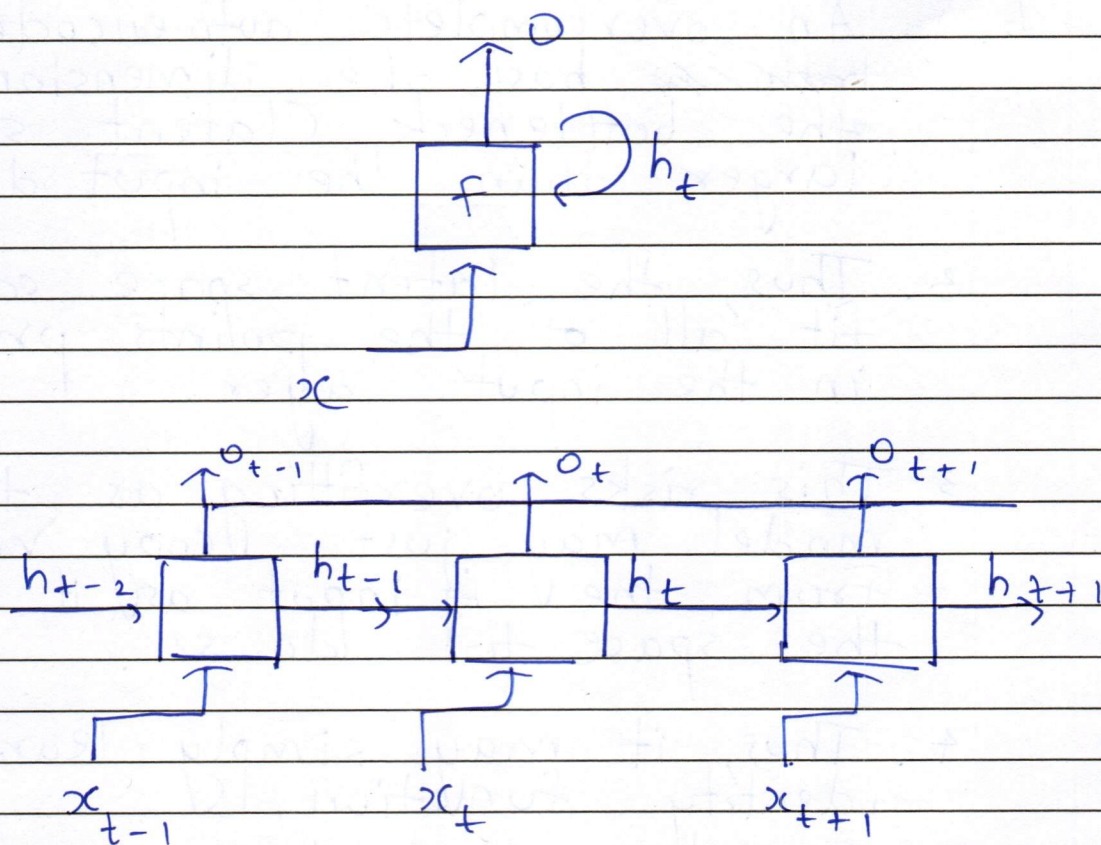
6. 1. An overcomplete autoencoder ~~train~~ has the dimensions of the bottleneck (latent space) larger than the input dimensions.
2. Thus, the latent space can fit all of the points present in the input layer.
3. This risks overfitting as the model may just copy values from the input as it has the space to do so.
4. Thus, it may simply learn the identity function.
5. Regularization ~~ex~~ adds a penalty term to the loss function.
6. This prevents the model from ~~lear~~ memorising input data.
7. Thus, without regularization ~~th~~ an over-complete autoencoder tends to learn the identity function.

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Q3. Architecture of RNN :

a.



Thus, the RNN loops over its hidden state. We derive the expression from the architecture :

$$h_t = W \cdot [h_{t-1}, x_t] + b_h$$

$$\therefore h_t = F(W_{hh} h_{t-1} + W_{xh} x_t + b_h) = L$$

$$\therefore \frac{\partial L}{\partial h_{t-k}} = \frac{\partial h_t}{\partial h_{t-k}} = F'(W_{hh} h_{t-1} + W_{xh} x_t + b_h)$$

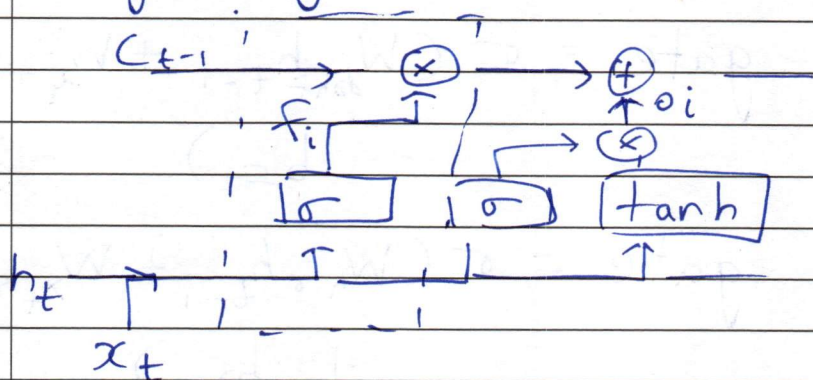
~~the function is usually a sigmoid function, with value between 0 and 1.~~

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Q3. When this term is multiplied with
a. learning rate η and by itself repeatedly due to its recursive nature, the value may become very very small over a longer period of time

forget gate in LSTM :



$$f_i = \sigma(x_t, h_t)$$

$$= \sigma(W_x x_t + W_h h_t + b)$$

The forget gate f_i is not directly multiplied by the hidden state. Instead, it is multiplied with previous cell state and then sent forward for further calculation. $[C_{t-1} * f_i]$

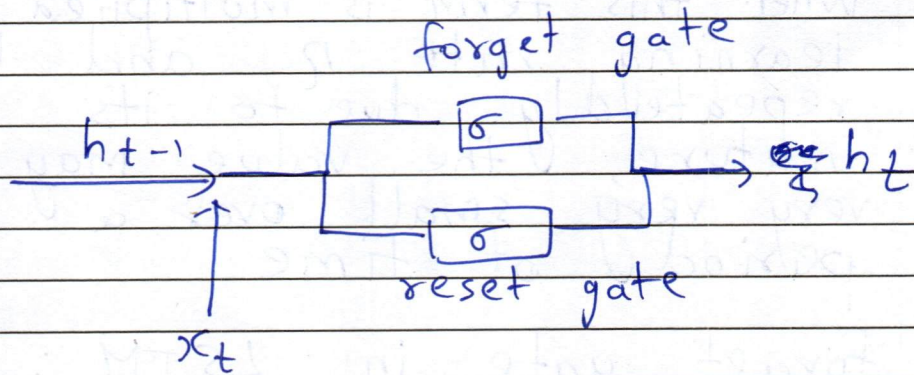
The final cell state is calculated as

$$C_t = C_{t-1} * f_i + o_i$$

Thus, since it is added not multiplied, the cell can choose whether to forget the information or not. This solves the problem of vanishing gradients due to its non multiplicative nature.

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Q3
6

$$\text{forget gate} = \sigma(W_{h_f} h_{t-1} + W_{x_f} x_t + b_f)$$

$$\text{reset gate} = \sigma(W_{h_r} h_{t-1} + W_{x_r} x_t + b_r)$$

Thus, instead of a separate cell state and hidden state, a GRU only has one state.

It also only has two gates - reset and update.

The input & forget

[illegible]

[illegible]